# Cluster 42

To view the latest Map of Science clusters and data, visit the live Map at sciencemap.eto.tech.

## Subjects

Research fields: Engineering, Materials science, Chemistry

The most common research fields of articles in this cluster. Read more.

Research subfields: Chemical engineering, Composite material, Nuclear chemistry

Common research subfields of articles in this cluster. Read more.

Key concepts: mechanical properties, polymer matrix, polymer matrix nanocomposites, scanning electron microscopy, MMT composite thermal

These key concepts are identified algorithmically using article titles and abstracts. Read more.

0.00% of all articles are AI-related

Articles are algorithmically labeled as AI-related or not. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to robotics

Articles are labeled as robotics-related or not using an ETO model. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to computer vision

Articles are labeled as computer vision-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to natural language processing

Articles are labeled as NLP-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to AI safety

Articles are labeled as related to AI safety or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to cybersecurity

## Vital signs

**269** recent articles in the cluster

269 articles in this cluster were published in the five year period ending in 2023.

Average article is **16.70** years old

The average age of articles in this cluster. However, most Map of Science metrics are calculated using articles from the past five years only.

Growth rating: **18.51th** percentile

Over the three year period ending in 2023, this cluster grew faster than roughly 18.51% of other clusters.

Citation rating: **45.88th** percentile

A cluster's citation rating is equal to the average citation percentile for all the articles in the cluster. Citation percentiles are calculated relative to articles of the same publication year.In this case, the average article in this cluster is cited more often than roughly 45.88% of all other articles published in the same year.

**89.96%** of articles are in English

**Not expected** to grow extremely fast

As determined by ETO's growth prediction model. Read more.

# Countries and languages

## Top author countries

Each row lists the number of articles in this cluster from the last five years with at least one author from an organization in the specified country. Data available for 227 of 269 articles. 143 of these articles had at least one author from an organization in these ten countries.

| Country | Articles (five year period ending in 2023) | Yearly citations per article (average) |
|---|---:|---:|
| India | 54 | 1.52 |
| China | 25 | 0.89 |
| United States | 14 | 1.75 |
| Saudi Arabia | 12 | 3.57 |
| Turkey | 12 | 0.93 |
| Russia | 11 | 0.64 |
| Japan | 9 | 2.94 |
| France | 8 | 1.54 |
| Iran | 8 | 0.81 |
| South Korea | 8 | 2.54 |

## Top funding countries

Data available for 49 of 269 articles. 32 of these articles disclosed funding from one or more organizations in these ten countries.

| Country | Articles (last five years) | Yearly citations (average) |
|---|---:|---:|
| India | 8 | 4.40 |
| South Korea | 5 | 3.09 |
| China | 5 | 1.63 |
| Brazil | 4 | 0.35 |
| Spain | 4 | 1.83 |
| Malaysia | 2 | 1.75 |
| Turkey | 2 | 1.54 |
| Belgium | 2 | 3.15 |
| Saudi Arabia | 2 | 0.80 |
| United States | 2 | 2.30 |

# Articles and sources

## Core articles

Core articles are especially highly connected to other papers within the cluster.

- An Investigative Study on the Progress of Nanoclay-Reinforced Polymers: Preparation, Properties, and Applications: A Review
  2021: Polymers. 74 citations.
- Mechanical, thermal, and morphological properties of low-density polyethylene nanocomposites reinforced with montmorillonite: Fabrication and characterizations
  2023: Cogent Engineering. 9 citations.
- Polysaccharide-Fibrous Clay Bionanocomposites and their Applications
  2022: Materials research foundations. 2 citations.
- The effect of montmorillonite modification and the use of coupling agent on mechanical properties of polypropylene–clay nanocomposites
  2020: Polymers and Polymer Composites. 11 citations.
- Mechanical and Moisture Barrier Properties of Epoxy–Nanoclay and Hybrid Epoxy–Nanoclay Glass Fibre Composites: A Review
  2022: Polymers. 17 citations.
- Coupled thermal stress and moisture absorption in modified a montmorillonite/copolymer nanocomposite: Experimental study
  2023: Polymer testing. 3 citations.
- State-of-the-Art Nanoclay Reinforcement in Green Polymeric Nanocomposite: From Design to New Opportunities
  2022: Minerals. 28 citations.

## Review articles

Review articles have between 100 and 1000 out-citations. These articles describe and systematize other contributors' research.

- DEVELOPMENT OF POLYURETHANE/CLAY NANOCOMPOSITES BASED ON PALM OIL POLYOL
  2023: Jurnal sains dan teknologi reaksi. 0 citations.
- An Investigative Study on the Progress of Nanoclay-Reinforced Polymers: Preparation, Properties, and Applications: A Review
  2021: Polymers. 74 citations.
- Epoxy Nanocomposites with Silicon-Based Nanomaterials
  2021: ACS symposium series. 2 citations.
- Polymer Nanocomposites
  2023: Eng—Advances in Engineering. 10 citations.
- Micro- and Nanoscale Structure Formation in Epoxy-Clay Nanocomposites
  2021: Epoxy Composites. 0 citations.
- Mechanical and Moisture Barrier Properties of Epoxy–Nanoclay and Hybrid Epoxy–Nanoclay Glass Fibre Composites: A Review
  2022: Polymers. 17 citations.
- Current synthesis and characterization techniques for clay-based polymer nano-composites and its biomedical applications: A review
  2022: Environmental research. 44 citations.
- Effect of melt blending processing on mechanical properties of polymer nanocomposites: a review
  2023: Polymer Bulletin. 12 citations.

## Highly cited articles

Highly cited articles are the articles in the cluster with the most citations overall.

- Physico-mechanical, rheological and gas barrier properties of organoclay and inorganic phyllosilicate reinforced thermoplastic films
  2020: Journal of Applied Polymer Science. 30 citations.
- An Investigative Study on the Progress of Nanoclay-Reinforced Polymers: Preparation, Properties, and Applications: A Review
  2021: Polymers. 74 citations.
- Effect of organo-smectite clays on the mechanical properties and thermal stability of EVA nanocomposites
  2020: Applied Clay Science. 25 citations.
- Single-Lap Joints Bonded with Epoxy Nanocomposite Adhesives: Effect of Organoclay Reinforcement on Adhesion and Fatigue Behaviors
  2021: ACS Applied Polymer Materials. 30 citations.
- Current synthesis and characterization techniques for clay-based polymer nano-composites and its biomedical applications: A review
  2022: Environmental research. 44 citations.
- Electrochemical and Mechanical Studies of Epoxy Coatings Containing Eco-Friendly Nanocomposite Consisting of Silane Functionalized Clay–Epoxy on Mild Steel
  2020: Journal of Bio- and Tribo-Corrosion. 26 citations.
- State-of-the-Art Nanoclay Reinforcement in Green Polymeric Nanocomposite: From Design to New Opportunities
  2022: Minerals. 28 citations.

## Top sources

The most common sources of articles from the last five years in this cluster, such as specific journals, conference proceedings, or preprint servers. Data available for 264 out of 269 articles. 50 of these articles came from these sources.

| Source | Articles (last five years) |
|---|---:|
| Polymers | 12 |
| Materials Today: Proceedings | 6 |
| Journal of Applied Polymer Science | 6 |
| Polymer Composites | 5 |
| Polymer Bulletin | 5 |
| AIP conference proceedings | 4 |
| Journal of Composites Science | 3 |
| Springer eBooks | 3 |
| The Egyptian Journal of Chemistry | 3 |
| International Journal of Polymer Analysis and Characterization | 3 |

# Authors, organizations, and funders

## Top authors

The authors with the most articles from the last five years in this cluster. Data available for 262 of 269 articles. 24 of these articles were written by these authors.

| Author | Articles (last five years) | Yearly citations (average) |
|---|---|---|
| Azlin Fazlina Osman | 5 | 2.22 |
| Oleksii Gonchar | 4 | 0.06 |
| V. A. Gerasin | 4 | 0.60 |
| В. А. Герасин | 4 | 0.43 |
| V. V. Kurenkov | 4 | 0.65 |
| Jin-Hae Chang | 4 | 5.33 |
| 陈斌 | 3 | 0.13 |
| Jin-Hae Chang | 3 | 3.56 |
| Zaleha Mustafa | 3 | 2.00 |
| M. Ramesh | 3 | 0.89 |

## Top author organizations

The author organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 232 of 269 articles. 34 had at least one author from one of these ten organizations.

| Organization | Articles (last five years) | Yearly citations (average) |
|---|---|---|
| King Saud University - Saudi Arabia | 5 | 0.92 |
| Indian Institute of Technology Madras - India | 4 | 0.17 |
| Universiti Malaysia Perlis - Malaysia | 4 | 1.2 |
| Centre National de la Recherche Scientifique - France | 4 | 1.58 |
| Institute of Macromolecular Chemistry - Ukraine | 4 | 0 |
| A.V. Topchiev Institute of Petrochemical Synthesis - Russia | 4 | 0.22 |
| Russian Academy of Sciences - Russia | 4 | 0.22 |
| University of Kufa - Iraq | 3 | 1.36 |
| Indian Institute of Technology Kharagpur - India | 3 | 3.56 |
| Aristotle University of Thessaloniki - Greece | 3 | 1 |

## Organization types

This table covers the 211 articles in this cluster from the five year period ending in 2023 with data about their author organization types (out of 269 articles total). Read more

| Commercial | Education | Nonprofit | Government |
|---|---|---|---|
| 0.72% | 45.60% | 10.25% | 4.18% |

## Top industry organizations

The industry organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 5 out of 269 articles. 5 of these articles had at least one author from an organization listed below.

| Organization | Articles (last five years) | Yearly citations (average) |
|---|---|---|
| Sonatrach (Algeria) | 1 | 0.6 |
| ShinEtsu Chemical (Japan) | 1 | 0 |
| Indian Oil Corporation (India) | 1 | 0.5 |
| Reliance Industries (India) | 1 | 0 |
| PPG Industries (United States) | 1 | 1 |

## Top funders

Data available for 79 out of 269 recent articles. 23 of these articles disclosed funding from one or more of these organizations.

| Funding organization | Articles (last five years) | Yearly citations (average) |
|---|---|---|
| National Natural Science Foundation of China | 4 | 1.88 |
| Council of Scientific & Industrial Research (CSIR) - India | 3 | 2.11 |
| European Union | 3 | 1.18 |
| Government of Spain | 3 | 1.89 |
| National Research Foundation of Korea - South Korea | 3 | 1.89 |
| National Research Foundation - South Korea | 3 | 3.56 |
| Ministry of Education, Science & Technological Development, Serbia | 2 | 1.57 |
| Ministry of Education - Malaysia | 2 | 1.75 |
| European Commission - Belgium | 2 | 3.15 |
| Ministry of Education - South Korea | 2 | 1.50 |

## Funder types

This table covers the 45 articles in this cluster from the last five years with data about their funding organization types (out of 269 articles total).

| Commercial | Education | Nonprofit | Government |
|---|---|---|---|
| 1.59% | 19.05% | 4.76% | 74.60% |

# Patents and industry

**722** patents cite articles in this cluster ( **98.27th** percentile among all clusters)

**6.64%** of articles in the cluster are cited by patents

**0.63%** of articles have at least one author from industry

## Top patent titles

| Patent title | Same-cluster citations |
|---|---|
| FLUORESCENT NANOCOMPOSITE MATERIALS BASED ON MODIFIED CLAY AND USES | 9 |
| SURFACTANT DERIVED FROM QUINOLINIUM FOR THE MODIFICATION OF CLAY | 9 |
| METHOD OF PREPARING A COMPOSITE WITH DISPERSE LONG FIBERS AND NANOPARTICLES | 34 |
| AN IMPROVED SOLUTION BLENDING PROCESS FOR THE FABRICATION OF NYLON6-MONTMORILLONITE NANOCOMPOSITES | 27 |
| METHOD FOR FORMING FLAME-RETARDANT CLAY-POLYOLEFIN COMPOSITES | 23 |
| ARTICLES COMPRISING A POLYIMIDE SOLVENT CAST FILM HAVING A LOW COEFFICIENT OF THERMAL EXPANSION AND METHOD OF MANUFACTURE THEREOF | 21 |
| METHOD FOR FORMING EXFOLIATED CLAY-POLYOLEFIN NANOCOMPOSITES | 15 |
| SPHERICAL PARTICLES COMPRISING NANOCLAY-FILLED-POLYMER AND METHODS OF PRODUCTION AND USES THEREOF | 11 |
| BLOCK COPOLYMER AND NANOFILLER COMPOSITES | 10 |

## Top patent assignees

| Patent assignee | Same-cluster citations |
|---|---|
| SATURN LICENSING, LLC | 40 |
| THE OHIO STATE UNIVERSITY | 37 |
| LEE L JAMES | 34 |
| ZHOU GANG | 34 |
| CAO XIA | 34 |
| AMERICAN UNIVERSITY IN CAIRO | 27 |
| UNIVERSITY OF BRUNEI DARUSSALAM | 27 |
| PEOPLES BRIAN | 23 |
| SCOTT SUSANNAH | 23 |

# Connections

## Top citing clusters

| Cluster | Number of significant citations | CSET extracted phrases |
|---|---|---|
| 7187 | 53 | Halloysite nanotubes, HNTs, Mechanical properties, nanocomposite films, adsorption |
| 24136 | 35 | PLA nanocomposites, PLA matrix, mechanical properties, neat PLA, Lactic Acid |
| 72708 | 33 | mechanical properties, glass fiber reinforced, fiber reinforced polymer, fiber epoxy composites, glass fiber |
| 5173 | 33 | maximum adsorption capacity, modified Bentonite, Montmorillonite, cation exchange capacity, clay minerals |
| 1612 | 28 | carbon nanotubes, mechanical properties, MWCNT nanocomposites, MWCNTs, multi-walled carbon nanotube |
| 14711 | 22 | polymer blends, polymer blend nanocomposites, mechanical properties, immiscible polymer blend, PLA |
| 414 | 22 | cellulose nanocrystals, CNCs, mechanical properties, acid hydrolysis, thermal stability |
| 31439 | 20 | drug release, drug delivery system, MMT, controlled drug delivery, chitosan |
| 16826 | 19 | Mechanical Properties, graphene nanoplatelets, graphene polymer nanocomposites, GNP loading, thermal stability |
| 4 | 18 | natural fiber composites, fiber reinforced polymer, mechanical properties, Hybrid Natural Fiber, reinforced polymer composite |

## Top cited clusters

| Cluster | Number of significant citations | CSET extracted phrases |
|---|---|---|
| 5173 | 16 | maximum adsorption capacity, modified Bentonite, Montmorillonite, cation exchange capacity, clay minerals |
| 414 | 13 | cellulose nanocrystals, CNCs, mechanical properties, acid hydrolysis, thermal stability |
| 24136 | 12 | PLA nanocomposites, PLA matrix, mechanical properties, neat PLA, Lactic Acid |
| 1612 | 12 | carbon nanotubes, mechanical properties, MWCNT nanocomposites, MWCNTs, multi-walled carbon nanotube |
| 4 | 9 | natural fiber composites, fiber reinforced polymer, mechanical properties, Hybrid Natural Fiber, reinforced polymer composite |
| 7187 | 9 | Halloysite nanotubes, HNTs, Mechanical properties, nanocomposite films, adsorption |
| 72708 | 7 | mechanical properties, glass fiber reinforced, fiber reinforced polymer, fiber epoxy composites, glass fiber |
| 20126 | 7 | epoxy resin, mechanical properties, epoxy composites, tensile strength, silica nanoparticles |
| 19624 | 7 | mechanical properties, epoxy composites, epoxy resin, GNPs, graphene nanoplatelets |
| 31439 | 6 | drug release, drug delivery system, MMT, controlled drug delivery, chitosan |

## Top GitHub repositories

No data available.